

```

In[13]:= (* All time and rate is normalized by  $\kappa$  *)
          |全部
          (* Number of span modes *)
          |数
dir = NotebookDirectory[];
          |当前笔记本的目录

SetDirectory[dir];
          |设置目录

hbar = 1.054 * 10-34;
Nspan = 60;
Ntot = 2 * Nspan + 1; (*Total number of modes*)
          |总计

(*Dispersion*)
d1a = (4.5665) * 1012; (*1550nm*)
d1b = 4.4189 * 1012; (*775nm*)
d1c = (4.25) * 1012; (*515nm*)

d2a = 2.0 * 8.84 * 108; (*1550nm*)
d2b = 2.0 * 4.31 * 108; (*775nm*)
d2c = 2.0 * 2.31 * 108; (*515nm*)

(* Mode frequency and dissipation rate *)

$$\omega_{0a} = 2\pi * \frac{3 * 10^8}{1550. * 10^{-9}};$$

nboff = 40.;
ncoff = 25.;
(* $\Omega_0 = 2\omega_0 + 0.0(\text{Random[]} - 0.5) * (d1 - D1) + 2(noff * d1 + noff^2 * d2 / 2) - (2noff * D1 + 4noff^2 * D2 / 2)$ *)
(*Here is also need to be checked maybe not good*)
          |当前位置

$$\omega_{0b} = 2\omega_{0a} + (nboff * d1a + nboff^2 * d2a / 2) - (nboff * d1b + nboff^2 * d2b / 2);$$


$$\omega_{0c} = 3\omega_{0a} + (ncoff * d1a + ncoff^2 * d2a / 2) - (ncoff * d1c + ncoff^2 * d2c / 2);$$


$$\omega_a[j_] := \omega_{0a} + d1a * j + \frac{d2a}{2.} * j^2;$$


$$\omega_b[j_] := \omega_{0b} + d1b * j + \frac{d2b}{2.} * j^2;$$


$$\omega_c[j_] := \omega_{0c} + d1c * j + \frac{d2c}{2.} * j^2;$$


$$\kappa_a = \frac{\omega_{0a}}{2 * 500. * 10^3};$$


$$\kappa_{a1} = \kappa_{a0} = \kappa_a / 2.;$$


$$\kappa_b = \frac{\omega_{0b}}{2 * 70. * 10^3};$$


$$\kappa_{b1} = \kappa_{b0} = \kappa_b / 2.;$$


$$\kappa_c = \frac{\omega_{0c}}{2 * 70. * 10^3};$$


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κc1 = κc0 = κc / 2.;
ωr = 2 π * 18.08 * 1012;
(*what is this r mode here?*) κr = 2 π * 60 * 109;

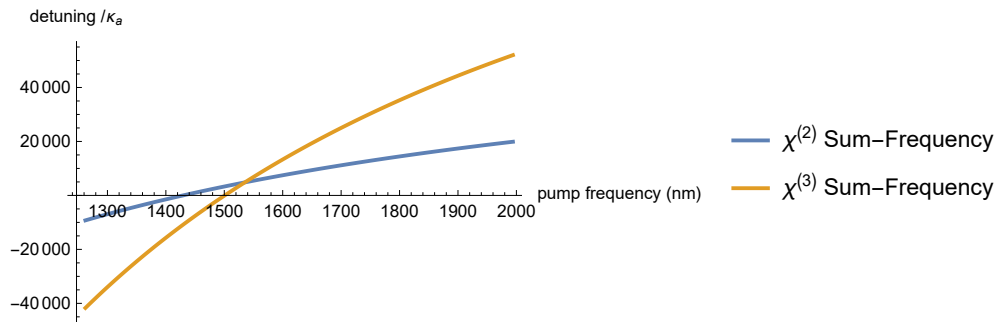
```

```
In[25]:= (*Plot the phase matching*)
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Print[ListLinePlot[Table[Table[ $\left\{\frac{2 \pi * 3. * 10^8}{\omega a[z]} * 10^9, \frac{\omega b[2 z] - 2 \omega a[z]}{\kappa a}\right\}$ , {z, -Nspan, Nspan}],
Table[ $\left\{\frac{2 \pi * 3. * 10^8}{\omega a[z]} * 10^9, \frac{\omega c[3 z] - 3 \omega a[z]}{\kappa a}\right\}$ , {z, -Nspan, Nspan}], PlotStyle →
Directive[Thick], PlotLegends → {"χ(2) Sum-Frequency", "χ(3) Sum-Frequency"},
AxesLabel → {"pump frequency (nm)", "detuning / κa"}];

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```
In[26]:= (* nonlinear coupling terms *)
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g2aab = 2 π * 0.08 * 1.0 * 106 * 1.00;
g2abc = 2 π * 0.08 * 1.0 * 106 * 0.10;
g3aa = -2 π * 2.8 * 1.00;
g3bb = -2 π * 14.9 * 1.00;
g3ab = -2 π * 17.1 * 1.00;
g3ac = -2 π * 0.1 * 1.00;
gR = 2 π * 0.0003 * 106;
(* pump to 1550 *)
p = 0; (* pump mode number *)
Pin = 1.0 * 2.9; (*Watt*) (*External Driving*)
ε =  $\sqrt{\frac{2 \kappa a1 * Pin}{\hbar * \omega 0a}}$ ; (*normalized driving amplitude*)

```

```

In[36]:= (*Initial cavity field*)
EA = EA0 = Table[{10-10-10*Random[]}, {i, 1, Ntot, 1}];
      |表格

EB = EC = ER = Table[{0.0}, {i, 1, Ntot, 1}]; (*why noise here*)
      |表格

(* Rearrange the frequency for FFT *)
liωa = liωb = liωc = {};
For[i = 1, i ≤ Ntot, i += 1,
  |For循环
    j = i - 1;
    If[i ≥ Nspan + 2, j = i - 1 - Ntot];
    |如果
    AppendTo[liωa, {ωa[j]}];
    |附加
    AppendTo[liωb, {ωb[j]}];
    |附加
    AppendTo[liωc, {ωc[j]}];
    |附加
  ];

In[40]:= (*Normalize Photon Number by*)
      |正规化      |数
Anorm = 1.0 * 103;
(* Normalize time by *)
      |正规化
κnorm = κa;

CombA = Table[{ $\frac{2 \pi * 3. * 10^8}{li\omega a[i, 1]} * 10^9$ ,
      |表格

    10 * Log[10, hbar * ω0a * 2 κa1 * Abs[EA[i, 1]]2 * Abs[Anorm]2 + 10-199], {i, 1, Ntot}];
      |对数

CombB = Table[{ $\frac{2 \pi * 3. * 10^8}{li\omega b[i, 1]} * 10^9$ ,
      |表格

    10 * Log[10, hbar * ω0b * 2 κb1 * Abs[EB[i, 1]]2 * Abs[Anorm]2 + 10-199], {i, 1, Ntot}];
      |对数

CombC = Table[{ $\frac{2 \pi * 3. * 10^8}{li\omega c[i, 1]} * 10^9$ ,
      |表格

    10 * Log[10, hbar * ω0c * 2 κc1 * Abs[EC[i, 1]]2 * Abs[Anorm]2 + 10-199], {i, 1, Ntot}];
      |对数

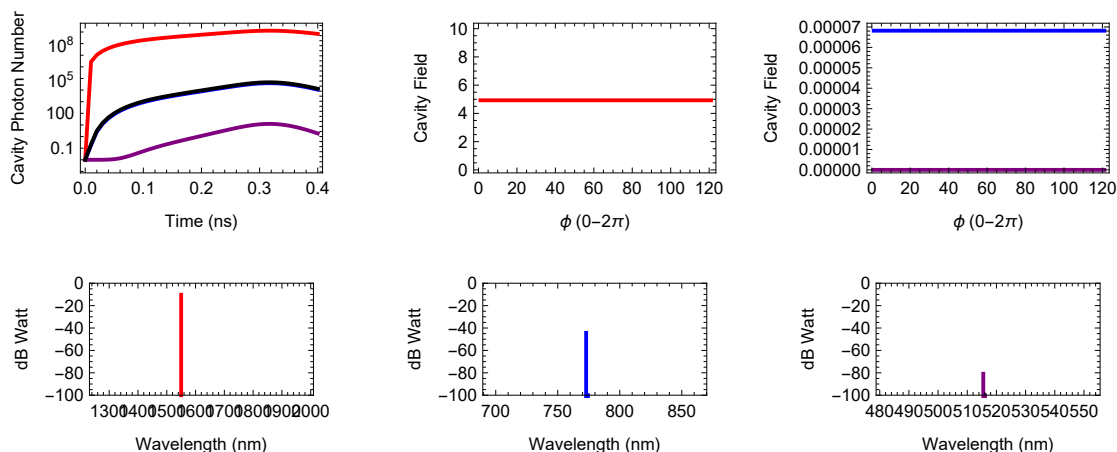
CombAav = CombBav = CombCav = Table[{0.0}, {i, 1, Ntot}];
      |表格

```

```

In[46]:= liNpA = liNpB = liNpC = liNpR = {};
Print[Dynamic[GraphicsGrid[
  |打印 |动态 |图形网格
  {{ListLogPlot[{liNpA, liNpB, liNpC, liNpR}, PlotRange → All, Joined → True,
    |点集的对数图 |绘制范围 |全部 |连接点 |真
    Axes → False, Frame → True, FrameLabel → {"Time (ns)", "Cavity Photon Number"},
    |坐标轴 |假 |边框 |真 |边框标签 |数
    PlotStyle → {Directive[Thick, Red], Directive[Thick, Blue],
    |绘图样式 |指令 |粗 |红色 |指令 |粗 |蓝色
    Directive[Thick, Purple], Directive[Thick, Black]}, ImageSize → 300},
    |指令 |粗 |紫色 |指令 |粗 |黑色 |图像尺寸
    ListLinePlot[{lica[[1]]}, PlotRange → All, Joined → True, Axes → False,
    |绘制点集的线条 |绘制范围 |全部 |连接点 |真 |坐标轴 |假
    Frame → True, FrameLabel → {" $\phi$  ( $0-2\pi$ )", "Cavity Field"},
    |边框 |真 |边框标签
    PlotStyle → {Directive[Thick, Red], Directive[Thick, Blue],
    |绘图样式 |指令 |粗 |红色 |指令 |粗 |蓝色
    Directive[Thick, Purple], Directive[Thick, Black]}, ImageSize → 300},
    |指令 |粗 |紫色 |指令 |粗 |黑色 |图像尺寸
    ListLinePlot[{licb[[1]], licc[[1]]}, PlotRange → All, Joined → True,
    |绘制点集的线条 |绘制范围 |全部 |连接点 |真
    Axes → False, Frame → True, FrameLabel → {" $\phi$  ( $0-2\pi$ )", "Cavity Field"},
    |坐标轴 |假 |边框 |真 |边框标签
    PlotStyle → {Directive[Thick, Blue], Directive[Thick, Purple],
    |绘图样式 |指令 |粗 |蓝色 |指令 |粗 |紫色
    Directive[Thick, Black]}, ImageSize → 300}],
    |指令 |粗 |黑色 |图像尺寸
    {ListPlot[CombA, PlotStyle → Red, Filling → -100, FillingStyle → Directive[
    |绘制点集 |绘图样式 |红色 |填补 |填充样式 |指令
    Thick, Red], ImageSize → 300, AspectRatio → 1 / 2, PlotRange → {All, {-100, 0}},
    |粗 |红色 |图像尺寸 |宽高比 |绘制范围 |全部
    Frame → True, FrameLabel → {"Wavelength (nm)", "dB Watt"}, Axes → False],
    |边框 |真 |边框标签 |坐标轴 |假
    ListPlot[CombB, PlotStyle → Blue, Filling → -200,
    |绘制点集 |绘图样式 |蓝色 |填补
    FillingStyle → Directive[Thick, Blue], ImageSize → 300,
    |填充样式 |指令 |粗 |蓝色 |图像尺寸
    AspectRatio → 1 / 2, PlotRange → {All, {-100, 0}}, Frame → True,
    |宽高比 |绘制范围 |全部 |边框 |真
    FrameLabel → {"Wavelength (nm)", "dB Watt"}, Axes → False], ListPlot[CombC,
    |边框标签 |坐标轴 |假 |绘制点集
    PlotStyle → Purple, Filling → -200, FillingStyle → Directive[Thick, Purple],
    |绘图样式 |紫色 |填补 |填充样式 |指令 |粗 |紫色
    ImageSize → 300, AspectRatio → 1 / 2, PlotRange → {All, {-100, 0}}, Frame → True,
    |图像尺寸 |宽高比 |绘制范围 |全部 |边框 |真
    FrameLabel → {"Wavelength (nm)", "dB Watt"}, Axes → False]]]]];
    |边框标签 |坐标轴 |假

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In[48]:=  $\delta\theta = -15.0;$ 
 $d\delta = 0.;$ 
(* $\delta\theta += d\delta * \frac{dt}{T}$ ; and what is this?*)
 $\delta = \delta\theta * \kappa a;$ 
 $\Delta a[j\_] := \frac{d2a}{2} * (j^2 - p^2) - \delta;$ 
(* $\Delta a[j\_] := \omega\theta a + d1a * j + \frac{d2a}{2} * j^2 - 1 \left( \omega\theta a + p * d1a + \frac{d2a}{2} * p^2 + \delta \right) - (j - p) d1a;$ *)
 $\Delta b[j\_] := \omega\theta b + d1b * j + \frac{d2b}{2} * j^2 - 2 \left( \omega\theta a + p * d1a + \frac{d2a}{2} * p^2 + \delta \right) - (j - 2p) d1a;$ 
 $\Delta c[j\_] := \omega\theta c + d1c * j + \frac{d2c}{2} * j^2 - 3 \left( \omega\theta a + p * d1a + \frac{d2a}{2} * p^2 + \delta \right) - (j - 3p) d1a;$ 
 $\Delta r[j\_] := \text{Sign}[\omega r - j * d1a] * \text{Min}[\text{Abs}[\omega r - j * d1a], 30 \kappa r];$ 

```

[正负符号] [绝对值]

```

In[55]:= (* Matrix of pump field *)
Pump = Table[{0.0}, {i, 1, Ntot, 1}];
      [表格]

Pump[[p + 1, 1]] =  $\frac{\epsilon}{\kappa_{\text{norm}} * \text{Anorm}}$ ;
Pump += EA0;

MCA =
      Table[{-i * Δa[i - 1 -  $\frac{1}{2} \times (1 + \text{Sign}[i - \text{Nspan} - 1.5]) \text{Ntot}$ ] - κa} /  $\kappa_{\text{norm}}$ , {i, 1, Ntot}];
      [表格] [正负符号]

MCb = Table[{-i * Δb[i - 1 -  $\frac{1}{2} \times (1 + \text{Sign}[i - \text{Nspan} - 1.5]) \text{Ntot}$ ] - κb} /  $\kappa_{\text{norm}}$ , {i, 1, Ntot}];
      [表格] [正负符号]

MCc = Table[{-i * Δc[i - 1 -  $\frac{1}{2} \times (1 + \text{Sign}[i - \text{Nspan} - 1.5]) \text{Ntot}$ ] - κc} /  $\kappa_{\text{norm}}$ , {i, 1, Ntot}];
      [表格] [正负符号]

MCR =
      Table[{-i * Δr[i - 1 -  $\frac{1}{2} \times (1 + \text{Sign}[i - \text{Nspan} - 1.5]) \text{Ntot}$ ] - κr} /  $\kappa_{\text{norm}}$ , {i, 1, Ntot}];
      [表格] [正负符号]

Idd = IdentityMatrix[Ntot];
      [单位矩阵]

MCI = Table[{1.0}, {i, 1, Ntot}];
      [表格]

G2aab = -2 i * g2aab *  $\sqrt{\text{Ntot}}$  *  $\frac{\text{Anorm}}{\kappa_{\text{norm}}}$  * MCI + EA0;

(*why we do the fourier transform here for the interaction parameters?*)

G2abc = -i * g2abc *  $\sqrt{\text{Ntot}}$  *  $\frac{\text{Anorm}}{\kappa_{\text{norm}}}$  * MCI + EA0;

G3aa = -2 i * g3aa * Ntot *  $\frac{\text{Anorm}^2}{\kappa_{\text{norm}}}$  * MCI + EA0;

G3ab = -i * g3ab * Ntot *  $\frac{\text{Anorm}^2}{\kappa_{\text{norm}}}$  * MCI + EA0;

G3bb = -2 i * g3bb * Ntot *  $\frac{\text{Anorm}^2}{\kappa_{\text{norm}}}$  * MCI + EA0;

G3ac = -i * g3ac * Ntot *  $\frac{\text{Anorm}^2}{\kappa_{\text{norm}}}$  * MCI + EA0;

GR = -i * gR *  $\sqrt{\text{Ntot}}$  *  $\frac{\text{Anorm}^2}{\kappa_{\text{norm}}}$  * MCI + EA0;

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In[69]:= (* FFT treatment ???*)
Pump = InverseFourier[Fourier[Pump]];
      |离散傅立叶逆变换 |傅立叶
G2aab = InverseFourier[Fourier[G2aab]];
      |离散傅立叶逆变换 |傅立叶
G2abc = InverseFourier[Fourier[G2abc]];
      |离散傅立叶逆变换 |傅立叶
G3aa = InverseFourier[Fourier[G3aa]];
      |离散傅立叶逆变换 |傅立叶
G3ab = InverseFourier[Fourier[G3ab]];
      |离散傅立叶逆变换 |傅立叶
G3bb = InverseFourier[Fourier[G3bb]];
      |离散傅立叶逆变换 |傅立叶
G3ac = InverseFourier[Fourier[G3ac]];
      |离散傅立叶逆变换 |傅立叶
GR = InverseFourier[Fourier[GR]];
      |离散傅立叶逆变换 |傅立叶
MCa = InverseFourier[Fourier[MCa]];
      |离散傅立叶逆变换 |傅立叶
MCb = InverseFourier[Fourier[MCb]];
      |离散傅立叶逆变换 |傅立叶

In[79]:= (*time scale*)
nanosec = 1.0 * 10-9;
picosec = 1.0 * 10-12;
T = 1 nanosec * xnorm;

In[82]:= (* Start the Simulation *)
lifig = {};
t1 = AbsoluteTime[] - 3 690 574 986;
      |绝对时间长度
Tglobal = 0.0;

(*time step*)
dt = 0.008 picosec * xnorm; Ndt = Round[0.01 T / dt];
      |舍入
PulseLength = 40.;
For[|iii = 0, |iii ≤ PulseLength * Ndt, |iii += Ndt,
|For循环
  (*Isb=Abs[A[[1+Nspan* 0+1,1]]2/e2;*)
      |绝对值
  IsA = Sum[Abs[EA[[kkkk, 1]]2, {kkkk, 1, Ntot}];
      |求和
  IsB = Sum[Abs[EB[[kkkk, 1]]2, {kkkk, 1, Ntot}];
      |求和
  IsC = Sum[Abs[EC[[kkkk, 1]]2, {kkkk, 1, Ntot}];
      |求和
  IsR = Sum[Abs[ER[[kkkk, 1]]2, {kkkk, 1, Ntot}];
      |求和
  If[IsA ≥ 1 000 000 000 000, |iii = 100 000 000; Break[]];
      |如里 |跳出循环

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AppendTo[linpA, {Tglobal, IsA * Anorm2 + 0.01}];
|附加

AppendTo[linpB, {Tglobal, IsB * Anorm2 + 0.01}];
|附加

AppendTo[linpC, {Tglobal, IsC * Anorm2 + 0.01}];
|附加

AppendTo[linpR, {Tglobal, IsR * Anorm2 + 0.01}];
|附加

For[jjj = 1, jjj ≤ Ndt, jjj += 1,
|For循环

  Tglobal +=  $\frac{dt}{xnorm} * 10^9$ ;

  (*1*)
  Fa = Fourier[EA]; Fb = Fourier[EB]; Fc = Fourier[EC]; Fr = Fourier[ER];
  |傅立叶 |傅立叶 |傅立叶 |傅立叶
  TWma = InverseFourier[Conjugate[Fa] * Fb];
  |离散傅立叶逆变换 |共轭
  TWmb = InverseFourier[Fa2];
  |离散傅立叶逆变换
  FWMaa = InverseFourier[Fa2 * Conjugate[Fa]];
  |离散傅立叶逆变换 |共轭
  FWMab = InverseFourier[Fa * Conjugate[Fb] * Fb];
  |离散傅立叶逆变换 |共轭
  FWMbb = InverseFourier[Fb2 * Conjugate[Fb]];
  |离散傅立叶逆变换 |共轭
  FWMba = InverseFourier[Fa * Conjugate[Fa] * Fb];
  |离散傅立叶逆变换 |共轭
  k1A = MCA * EA + G2aab * TWma + G2abc * InverseFourier[Conjugate[Fb] * Fc] +
  |离散傅立叶逆变换 |共轭
  G3aa * FWMaa + G3ab * FWMab + G3ac * InverseFourier[Conjugate[Fa]2 * Fc] +
  |离散傅立叶逆变换
  GR * InverseFourier[Fa * Fr] + GR * InverseFourier[Fa * Conjugate[Fr]] + Pump;
  |离散傅立叶逆变换 |离散傅立叶逆变换 |共轭
  k1B = MCB * EB + G2aab * TWmb + G2abc * InverseFourier[Conjugate[Fa] * Fc] +
  |离散傅立叶逆变换 |共轭
  G3bb * FWMbb + G3ab * FWMba;
  k1C = MCC * EC + G2abc * InverseFourier[Fa * Fb] + G3ac * InverseFourier[Fa3];
  |离散傅立叶逆变换 |离散傅立叶逆变换
  k1R = MCR * ER + GR * InverseFourier[Conjugate[Fa] * Fa];
  |离散傅立叶逆变换 |共轭

  (*2*)
  EAp = EA +  $\frac{dt}{2} * k1A$ ;
  EBp = EB +  $\frac{dt}{2} * k1B$ ;

```



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ECp = EC +  $\frac{dt}{2}$  * k1C;
ERp = ER +  $\frac{dt}{2}$  * k1R;
Fa = Fourier[EAp];
    傅立叶
Fb = Fourier[EBp];
    傅立叶
Fc = Fourier[ECp];
    傅立叶
Fr = Fourier[ERp];
    傅立叶
TWMa = InverseFourier[Conjugate[Fa] * Fb];
    离散傅立叶逆变换 共轭
TWMb = InverseFourier[Fa2];
    离散傅立叶逆变换
FWMa = InverseFourier[Fa2 * Conjugate[Fa]];
    离散傅立叶逆变换 共轭
FWMb = InverseFourier[Fa * Conjugate[Fb] * Fb];
    离散傅立叶逆变换 共轭
FWMbb = InverseFourier[Fb2 * Conjugate[Fb]];
    离散傅立叶逆变换 共轭
FWMba = InverseFourier[Fa * Conjugate[Fa] * Fb];
    离散傅立叶逆变换 共轭
k2A = MCa * EAp + G2aab * TWMa + G2abc * InverseFourier[Conjugate[Fb] * Fc] +
    离散傅立叶逆变换 共轭
    G3aa * FWMaa + G3ab * FWMab + G3ac * InverseFourier[Conjugate[Fa]2 * Fc] +
    离散傅立叶逆变换
    GR * InverseFourier[Fa * Fr] + GR * InverseFourier[Fa * Conjugate[Fr]] + Pump;
    离散傅立叶逆变换 离散傅立叶逆变换 共轭
k2B = MCb * EBp + G2aab * TWMb + G2abc * InverseFourier[Conjugate[Fa] * Fc] +
    离散傅立叶逆变换 共轭
    G3bb * FWMbb + G3ab * FWMba;
k2C = MCC * ECp + G2abc * InverseFourier[Fa * Fb] + G3ac * InverseFourier[Fa3];
    离散傅立叶逆变换 离散傅立叶逆变换
k2R = MCr * ERp + GR * InverseFourier[Conjugate[Fa] * Fa];
    离散傅立叶逆变换 共轭

(*3*)
EAp = EA +  $\frac{dt}{2}$  * k2A;
EBp = EB +  $\frac{dt}{2}$  * k2B;
ECp = EC +  $\frac{dt}{2}$  * k2C;
ERp = ER +  $\frac{dt}{2}$  * k2R;
Fa = Fourier[EAp];
    傅立叶

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```

Fb = Fourier[EBp];
    傅立叶
Fc = Fourier[ECp];
    傅立叶
Fr = Fourier[ERp];
    傅立叶
TWMa = InverseFourier[Conjugate[Fa] * Fb];
    离散傅立叶逆变换 共轭
TWMb = InverseFourier[Fa^2];
    离散傅立叶逆变换
FWMaa = InverseFourier[Fa^2 * Conjugate[Fa]];
    离散傅立叶逆变换 共轭
FWMab = InverseFourier[Fa * Conjugate[Fb] * Fb];
    离散傅立叶逆变换 共轭
FWMbb = InverseFourier[Fb^2 * Conjugate[Fb]];
    离散傅立叶逆变换 共轭
FWMba = InverseFourier[Fa * Conjugate[Fa] * Fb];
    离散傅立叶逆变换 共轭
k3A = MCa * EAp + G2aab * TWMa + G2abc * InverseFourier[Conjugate[Fb] * Fc] +
    离散傅立叶逆变换 共轭
    G3aa * FWMaa + G3ab * FWMab + G3ac * InverseFourier[Conjugate[Fa]^2 * Fc] +
    离散傅立叶逆变换
    GR * InverseFourier[Fa * Fr] + GR * InverseFourier[Fa * Conjugate[Fr]] + Pump;
    离散傅立叶逆变换 离散傅立叶逆变换 共轭
k3B = MCb * EBp + G2aab * TWMb + G2abc * InverseFourier[Conjugate[Fa] * Fc] +
    离散傅立叶逆变换 共轭
    G3bb * FWMbb + G3ab * FWMba;
k3C = MCc * ECp + G2abc * InverseFourier[Fa * Fb] + G3ac * InverseFourier[Fa^3];
    离散傅立叶逆变换 离散傅立叶逆变换
k3R = MCr * ERp + GR * InverseFourier[Conjugate[Fa] * Fa];
    离散傅立叶逆变换 共轭

(*4*)
EAp = EA + dt * k3A;
EBp = EB + dt * k3B;
ECp = EC + dt * k3C;
ERp = ER + dt * k3R;
Fa = Fourier[EAp];
    傅立叶
Fb = Fourier[EBp];
    傅立叶
Fc = Fourier[ECp];
    傅立叶
Fr = Fourier[ERp];
    傅立叶
TWMa = InverseFourier[Conjugate[Fa] * Fb];
    离散傅立叶逆变换 共轭
TWMb = InverseFourier[Fa^2];
    离散傅立叶逆变换
FWMaa = InverseFourier[Fa^2 * Conjugate[Fa]];
    离散傅立叶逆变换 共轭

```

```

FWMab = InverseFourier[Fa * Conjugate[Fb] * Fb];
      |离散傅立叶逆变换      |共轭
FWMbb = InverseFourier[Fb^2 * Conjugate[Fb]];
      |离散傅立叶逆变换      |共轭
FWMba = InverseFourier[Fa * Conjugate[Fa] * Fb];
      |离散傅立叶逆变换      |共轭
k4A = MCa * EAp + G2aab * TWMa + G2abc * InverseFourier[Conjugate[Fb] * Fc] +
      |离散傅立叶逆变换      |共轭
      G3aa * FWMaa + G3ab * FWMab + G3ac * InverseFourier[Conjugate[Fa]^2 * Fc] +
      |离散傅立叶逆变换
      GR * InverseFourier[Fa * Fr] + GR * InverseFourier[Fa * Conjugate[Fr]] + Pump;
      |离散傅立叶逆变换      |离散傅立叶逆变换      |共轭
k4B = MCb * EBp + G2aab * TWMb + G2abc * InverseFourier[Conjugate[Fa] * Fc] +
      |离散傅立叶逆变换      |共轭
      G3bb * FWMbb + G3ab * FWMba;
k4C = MCc * ECp + G2abc * InverseFourier[Fa * Fb] + G3ac * InverseFourier[Fa^3];
      |离散傅立叶逆变换      |离散傅立叶逆变换
k4R = MCr * ERp + GR * InverseFourier[Conjugate[Fa] * Fa];
      |离散傅立叶逆变换      |共轭

(*end*)
1
EA += - * dt * (k1A + 2 k2A + 2 k3A + k4A);
6
1
EB += - * dt * (k1B + 2 k2B + 2 k3B + k4B);
6
1
EC += - * dt * (k1C + 2 k2C + 2 k3C + k4C);
6
1
ER += - * dt * (k1R + 2 k2R + 2 k3R + k4R)
6
];
lica = Abs[Transpose[Fa]]^2;
licb = Abs[Transpose[Fb]]^2;
licc = Abs[Transpose[Fc]]^2;

CombAav += EA * Conjugate[EA];
      |共轭
CombBav += EB * Conjugate[EB];
      |共轭
CombCav += EC * Conjugate[EC];
      |共轭

CombA = Table[{{
      |表格
      2 * Pi * 3. * 10^8
      ----- * 10^9,
      liwa[[i, 1]]

      10 * Log[10, hbar * omega0a * 2 ka1 * Abs[EA[[i, 1]]]^2 * Abs[Anorm^2] + 10^-199]}, {i, 1, Ntot}];
      |对数      |绝对值

CombB = Table[{{
      |表格
      2 * Pi * 3. * 10^8
      ----- * 10^9, 10 * Log[10,
      |对数

```

```

hbar * ω0b * 2 κb1 * Abs[EB[[i, 1]]2 * Abs[Anorm2 + 10-199]]}, {i, 1, Ntot}];
|绝对值

CombC = Table[ {  $\frac{2 \pi * 3. * 10^8}{1 i \omega c[[i, 1]]} * 10^9$ , 10 * Log[10,
|表格 |对数

hbar * ω0c * 2 κb1 * Abs[EC[[i, 1]]2 * Abs[Anorm2 + 10-199]]}, {i, 1, Ntot} ]
|绝对值

];

t2 = AbsoluteTime[] - 3 690 574 986;
|绝对时间长度

Print["Time used: ", t2 - t1, " Seconds"];
|打印

Time used: 213.2281650 Seconds

In[90]:= PBtot = Sum[hbar * ω0b * 2 κb1 * Abs[Anorm]2 * CombBav[[i, 1]] / PulseLength, {i, 1, Ntot}];
|求和

PCtot = Sum[hbar * ω0c * 2 κb1 * Abs[Anorm]2 * CombCav[[i, 1]] / PulseLength, {i, 1, Ntot}];
|求和

PATot = Sum[hbar * ω0a * 2 κa1 * Abs[Anorm]2 * CombAav[[i, 1]] / PulseLength, {i, 1, Nspan}] +
|求和
Sum[hbar * ω0a * 2 κa1 * Abs[Anorm]2 * CombAav[[i, 1]] / PulseLength, {i, 2 + Nspan, Ntot}];
|求和

Print["The power of A,B,C are ", Re[PATot], "Watt, ",
|打印 |常量 |实部
Re[PBtot], "Watt", Re[PCtot], "Watt"];
|实部 |实部

The power of A,B,C are 0.0992268Watt, 0.0000607503Watt2.30549×10-8Watt

```

```

In[94]:= CombAavout = Table[ $\left\{\frac{2 \pi * 3. * 10^8}{li\omega a[i, 1]} * 10^9,$ 
      [表格]
      10 * Log[10, hbar *  $\omega\theta a * 2 \kappa a1 * Abs[Anorm^2]$  * CombAav[i, 1] / PulseLength +  $10^{-199}$ ], {i,
      [对数] [绝对值]
      1, Ntot}];

CombBavout = Table[ $\left\{\frac{2 \pi * 3. * 10^8}{li\omega b[i, 1]} * 10^9,$ 
      [表格]
      10 * Log[10, hbar *  $\omega\theta b * 2 \kappa b1 * Abs[Anorm^2]$  * CombBav[i, 1] / PulseLength +  $10^{-199}$ ], {i,
      [对数] [绝对值]
      1, Ntot}];

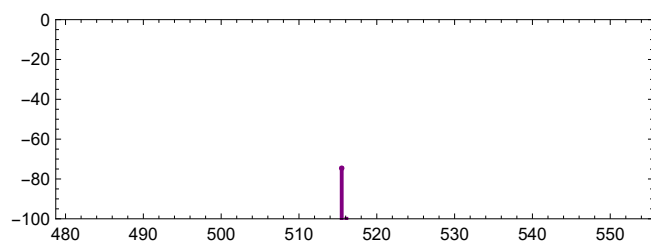
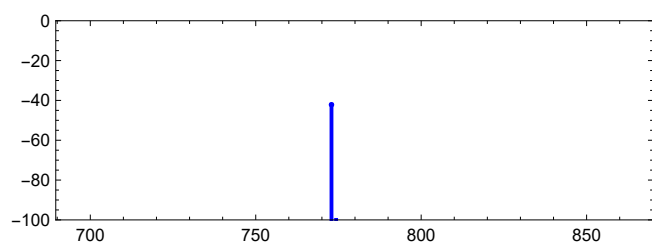
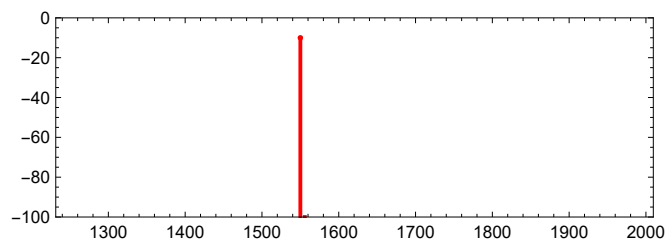
CombCavout = Table[ $\left\{\frac{2 \pi * 3. * 10^8}{li\omega c[i, 1]} * 10^9,$ 
      [表格]
      10 * Log[10, hbar *  $\omega\theta c * 2 \kappa c1 * Abs[Anorm^2]$  * CombCav[i, 1] / PulseLength +  $10^{-199}$ ], {i,
      [对数] [绝对值]
      1, Ntot}];

figav = GraphicsGrid[{{ListPlot[CombAavout, PlotStyle → Red, Filling → -200,
      [图形网格] [绘制点集] [绘图样式] [红色] [填补]
      FillingStyle → Directive[Thick, Red], ImageSize → 400, AspectRatio → 1 / 3,
      [填充样式] [指令] [粗] [红色] [图像尺寸] [宽高比]
      PlotRange → {All, {-100, 0}}, Frame → True, Axes → False]},
      [绘制范围] [全部] [边框] [真] [坐标轴] [假]
      {ListPlot[CombBavout, PlotStyle → Blue, Filling → -200,
      [绘制点集] [绘图样式] [蓝色] [填补]
      FillingStyle → Directive[Thick, Blue], ImageSize → 400, AspectRatio → 1 / 3,
      [填充样式] [指令] [粗] [蓝色] [图像尺寸] [宽高比]
      PlotRange → {All, {-100, 0}}, Frame → True, Axes → False]},
      [绘制范围] [全部] [边框] [真] [坐标轴] [假]
      {ListPlot[CombCavout, PlotStyle → Purple, Filling → -200,
      [绘制点集] [绘图样式] [紫色] [填补]
      FillingStyle → Directive[Thick, Purple], ImageSize → 400, AspectRatio → 1 / 3,
      [填充样式] [指令] [粗] [紫色] [图像尺寸] [宽高比]
      PlotRange → {All, {-100, 0}}, Frame → True, Axes → False}}];
      [绘制范围] [全部] [边框] [真] [坐标轴] [假]

```

In[98]:= **Print[figav];**

[打印](#)



In[99]:= **(*rig=ToString[Round[Random[]*1000000]]];*)**

[\[转换为...\] \[舍入\]](#)

rig = ToString[kkkk];

[\[转换为字符串\]](#)

Mname = \$MachineName;

[\[机器名称\]](#)

lidade = DateList[];

[\[日期列表\]](#)

timeStr = ToString[lidade[[1]]] <> ToString[lidade[[2]]] <>

[\[转换为字符串\]](#)

[\[转换为字符串\]](#)

ToString[lidade[[3]]] <> ToString[lidade[[4]]] <> ToString[lidade[[5]]];

[\[转换为字符串\]](#)

[\[转换为字符串\]](#)

[\[转换为字符串\]](#)

nb = InputNotebook[];

[\[输入笔记本\]](#)

NotebookPrint[nb, dir <> rig <> "-" <> Mname <> "-" <> timeStr <> ".pdf"];

[\[笔记本打印\]](#)

Export[rig <> "-" <> Mname <> "-" <> timeStr <> "-average.png", figav]

[\[导出\]](#)

Out[99]= kkkk-laptop-ufk6inm7-20232121620-average.png